

AI ‘know’ their inputs—self-aware systems. When we started, there was a lot of buzz around this typical approach to AI.

We don’t follow a black box approach towards intelligence; there is no such ‘one monolithic system.’ Cybernetics, in fact, precedes AI as a structure. We often confuse intelligence with the ability to perform phenomenal tasks or handle sophisticated things. However, according to the field of cybernetics, intelligence is the outcome and not the source or cause. Further, intelligence is the ability of a system to self-regulate and handle a sophisticated situation.

Remember those enormously large picturesque flock of birds moving in unison like one intelligent organism, forming extremely complex patterns? Or the large schools of fish that swiftly turn together in a complex way to confuse the predators, like a single intelligent being. Not every fish or bird saw the predator. They just loosely followed the one ahead of them, and yet they end up achieving a very complex behaviour. Ants, locusts, and many other living beings exhibit this collective behaviour which is not just swarm intelligence but more fundamental to even seemingly non-swarm organisms like the human body.

A cell in the liver doesn’t realise that it is reacting to a command from a neuron in the brain, and neither does the neuron in the brain realise that it’s sending a command to a liver. Even a neuron doesn’t realise that it’s reacting to another neuron. And here we are—two bundles of such simple cells curiously discussing all this!

Behind every such complex and intelligent behaviour there are simple systems just adamantly trying to achieve a primitive fixed task. Interestingly, when you put those simple systems with each other, that adamancy forces them to rearrange and constantly adjust to each other’s stimuli. As a result they

end up solving a complex situation, without any awareness of the bigger picture, creating an appearance/illusion of intelligence.

This self-regulatory behaviour is Cybernetics.

If you look at your brain, the part that handles sight keeps doing image processing, whereas the audio portion does certain kinds of audio processing. When they come in tandem with each other, there’s a

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sophisticated handling of our reaction to sound and sight. This coming together of these two faculties gives meaning to each other. The moment I say orange, you are able to visualise the picture of an orange or the orange colour, and this gives the meaning to the sound of “orange.” This is vision. This approach is called Cybernetics—the ability of a system to be self-regulatory.

Nikhil: The second part of our name comes from labour. Most people don’t know this. A laboratory literally means a place where labour happens. The meaning has changed over time. But basically, robots are an alternative to human labour. And that’s why the name is *Cybernetics Laboratory*.

You develop industrial robots or, to be precise, visual object intelligence for industrial robots. What exactly is meant by the term ‘visual object intelligence?’

Gokul: Unlike every other technology, vision has kind of become

over-assumed and vocabulary-insufficient. When we have to make a system go and interact with an object, like when a robotic arm in an industrial setting needs to pick up an object, it first needs to be able to recognise the object visually. But we tend to limit vision to colour and depth. An object that has so many features like weight, shape, and texture gets reduced and abstracted down to only a colour pattern. But that’s not enough for the robots to successfully manipulate objects. That is the limitation that we are trying to solve.

Another common misconception is the confusion between sight and vision. Let’s say I give you an object in your hand. You have never seen this object before. You don’t know what it is, you’re not able to identify what it is. You pick up that object and start tilting it, rotating it, and manipulating it. So, you have manipulated that object even before you know what this object is. This is where we confuse sight with vision. Sight gives you a certain amount of data on a 3D object, but vision gives meaning to that data.

Like I mentioned before, when I say the word orange, you start imagining all the other feelings with your associated senses—the texture, smell, taste (you might start salivating). All these associated feelings give meaning to the object. All that comes together only with imagination, the holistic understanding of that object being associated with every other sensory feeling that’s needed for you to be able to interact with it. That is intelligence, and that’s when vision actually happens. That’s what we mean by visual object intelligence.

This intelligence, when available in robots, allows them to grasp any object without any prior training. Because of this, robots can learn to operate items in any orientation, in addition to grasping objects in unstructured environments.